Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI)

Project Scope Document

for

Project ABC: Smart Irrigation System for Small Local Farms

Version 1.0

by

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Bachelor of Science in Artificial Intelligence (2022-2026)

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Abstract

Water scarcity and inefficient water management present critical challenges for small-scale farmers in many regions [Optional: You could add 'like Khyber Pakhtunkhwa'], often leading to suboptimal crop yields, significant water wastage, and increased operational costs. Current irrigation practices frequently rely on manual estimations or fixed schedules, which fail to adapt to real-time variations in soil moisture, specific crop needs, or fluctuating weather conditions. This often results in either under-watering, stressing crops and reducing yield, or over-watering, which wastes precious water resources, leaches nutrients, and increases energy consumption for pumping. Project ABC bridges this gap by developing an affordable and automated Smart Irrigation System specifically designed for the needs of local small farms. The system utilizes a network of soil moisture sensors placed strategically in the field, combined with local weather data [Optional: specify source, e.g., 'fetched online' or 'from a simple local sensor'], to determine precise watering requirements. A microcontroller-based control unit processes this data and automatically activates the irrigation system (e.g., water pump, valves) only when and where needed, delivering the optimal amount of water directly to the root zone. By replacing guesswork with data-driven automation, Project ABC aims to significantly conserve water resources, reduce electricity/fuel costs associated with pumping, minimize manual labor, and ultimately improve crop health and overall farm productivity, promoting more sustainable and resilient farming practices..

1. Introduction

Agriculture, particularly smallholder farming, forms the backbone of the economy and livelihoods in regions like Topi, Khyber Pakhtunkhwa. Effective water management is crucial for farm productivity, yet traditional irrigation methods are increasingly strained by water scarcity and climate variability. As a community reliant on agriculture, we must ensure that local farmers have access to efficient and sustainable tools to manage water resources effectively. It is important for farmers to have a system that provides automated, needs-based irrigation, minimizing waste and maximizing crop potential while reducing manual labor.

Traditional irrigation practices in our area often rely on manual control or fixed timers, leading to significant water wastage, inconsistent soil moisture, and potential crop damage. Unlike these conventional methods, Project ABC, a proposed Smart Irrigation System, incorporates sensor technology and automated controls to optimize water usage based on real-time conditions. Traditional approaches lack the ability to respond dynamically to actual soil moisture levels or changing weather patterns, creating unnecessary burdens on farmers and straining limited water and energy resources. By combining soil moisture sensing, localized weather data consideration [Optional: could specify source later, e.g., 'via simple sensors or online forecasts'], and automated water delivery control, Project ABC offers a practical, cost-effective solution to help small local farmers manage irrigation efficiently and sustainably.

2. Problem Statement

In the agricultural landscape of Topi and surrounding areas in Khyber Pakhtunkhwa, small-scale farmers face significant challenges in managing water for irrigation. Limited water availability, coupled with increasingly unpredictable rainfall patterns due to climate change (as observed globally and impacting our region), makes efficient water use paramount. The existing methods predominantly involve manual irrigation (flooding or hose watering) or basic timers, which are often inefficient. These methods lead to substantial over-watering, wasting precious water and energy (for pumps), or under-watering, stressing crops and reducing yields. Farmers often rely on guesswork or fixed schedules that don't account for the actual moisture needs of the soil and crops, which vary daily based on weather and plant growth stages. This inefficiency not only impacts crop health and farm profitability but also contributes to environmental issues like water resource depletion and potential soil salinity buildup from excessive watering. Furthermore, manual irrigation is labor-intensive, demanding significant time and effort from farmers who often juggle multiple responsibilities. Existing automated solutions are frequently expensive or too complex for the context of local smallholders, lacking the affordability and simplicity needed for widespread adoption in our community.

3. Problem Solution/Objectives of the Proposed System

There is a clear need for a system that optimizes irrigation for small local farms, ensuring water conservation, resource efficiency, and improved crop outcomes. This system would allow farmers to automate watering based on actual environmental data, reducing reliance on manual labor and guesswork. It would utilize low-cost sensors to monitor soil conditions and potentially factor in local weather forecasts to make intelligent irrigation decisions. By integrating automated control of water pumps or valves, the platform would ensure timely and precise water delivery directly where needed. Additionally, features such as simple status monitoring (e.g., via an LCD or basic alerts) and robust, low-maintenance hardware would enhance usability and reliability for local farmers. This system would provide a practical, accessible solution for smart water management, eliminating inefficiencies and allowing farmers to focus on other critical aspects of farming while promoting sustainability.

3.1 Objectives

BO-1: Reduce water consumption for irrigation compared to traditional manual methods by automatically watering only when soil moisture drops below a set threshold.

BO-2: Improve crop health and potential yield by maintaining soil moisture within an optimal range suitable for common local crops.

BO-3: Decrease farmer labor time and effort associated with manual irrigation scheduling and operation.

BO-4: Develop a cost-effective prototype using readily available and affordable components suitable for smallholder farmers in the local context.

BO-5: Ensure system reliability and durability under typical local environmental conditions (e.g., dust, temperature fluctuations).

BO-6: Provide a simple and intuitive interface for farmers to monitor system status and potentially adjust basic settings (e.g., moisture threshold).

BO-7: Minimize energy consumption by activating water pumps only when necessary and for the required duration..

4. Related System Analysis/Literature Review

Table 1 : Related System Analysis with proposed project solution

Application Name	**Features**	**Weakness**	**Relevance with the Proposed System – How these weaknesses are addressed**
System-1	• Feature 1 • Feature 2	• Weakness 1 • Weakness 2	• Statement 1 • Statement 2

5. Vision Statement

For small local farmers struggling with unpredictable weather patterns, water scarcity, and the labor-intensive nature of traditional irrigation, FarmFlow is an accessible, affordable smart irrigation system that will provide automated, data-driven watering schedules, real-time soil moisture monitoring, and remote management capabilities, allowing farmers to optimize water usage and improve crop yields efficiently. Unlike conventional irrigation methods that lead to water waste, inconsistent crop growth, and significant manual effort, our system will ensure precise watering, conserve water resources, and reduce labor costs, enabling farmers to achieve more sustainable and profitable farming while contributing to environmental conservation.

6. Scope

FarmFlow is a smart irrigation system specifically designed to address the unique needs and challenges of small local farms. The system aims to streamline irrigation practices, ensuring farmers can manage their water resources effectively, reduce operational costs, and enhance crop productivity. The core of FarmFlow lies in its automated, data-driven approach. The system will utilize a network of affordable soil moisture sensors strategically placed throughout the farm to collect real-time data on soil hydration levels. This data, combined with local weather forecasts obtained through integration with publicly available weather services, will feed into an intelligent algorithm that calculates the optimal watering schedule for different crops and areas of the farm. The platform will provide farmers with an intuitive mobile application and a web interface for monitoring soil conditions, viewing upcoming irrigation schedules, and receiving alerts. Farmers will have the ability to remotely adjust settings and manually trigger or pause irrigation cycles via the application.

Key features within the scope include:

- Automated Scheduling: Dynamically adjusts irrigation timing and duration based on real-time soil moisture data and weather forecasts.
- Soil Moisture Monitoring: Provides farmers with granular insights into soil hydration levels across their fields.
- Weather Data Integration: Incorporates local weather information (rainfall, temperature, humidity) to optimize watering and prevent unnecessary irrigation.
- Remote Monitoring and Control: Allows farmers to access system data and control irrigation remotely via a mobile or web application.
- Alerts and Notifications: Notifies farmers about critical events such as low soil moisture levels, system malfunctions, or significant weather changes.
- Basic Reporting: Provides simple reports on water usage over time and system activity.

The system will be designed for easy installation and integration with common small-scale irrigation infrastructure (e.g., existing pumps and basic piping). The initial scope will focus on field crops and common irrigation methods suitable for small farms, such as drip irrigation and basic sprinkler systems..

7. Project Stakeholders and Roles

Table 2: Project Stakeholders for [PROJECT]

Project Sponsor	[GIKI]
Stakeholder	• [PROJECT MEMBER] • [PROJECT SUPERVISOR] • Senior Design Project Evaluation Panel

8. References

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